



Bridging Standards and Deployments

At the intersection of Media and Connectivity, for generations to come

Frédéric Gabin
5G-MAG Chair

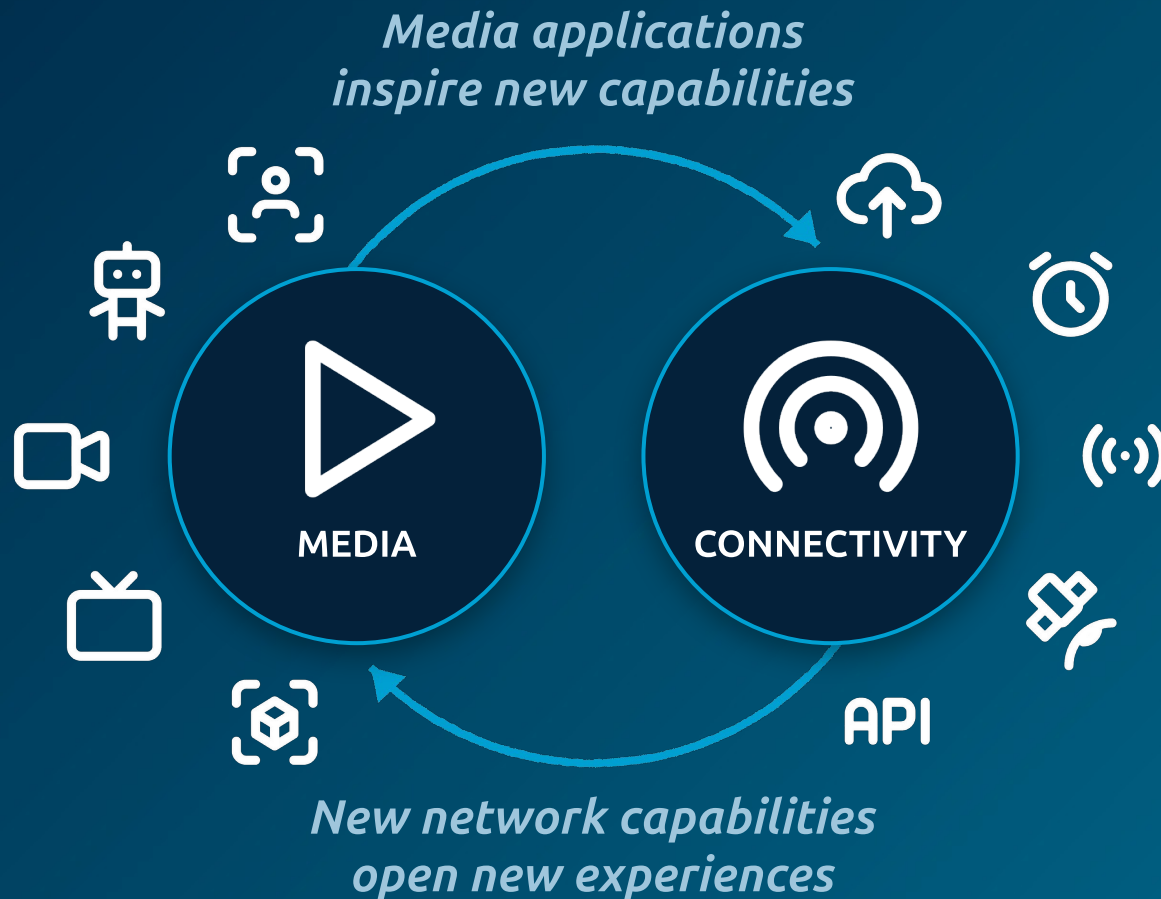


**STREAMING – PRODUCTION – BROADCAST – REAL-TIME COMMS – AVATAR – EXTENDED REALITY
5G/6G – NON-TERRESTRIAL NETWORKS – DATA COLLECTION – NETWORK APIs – COMPUTING – AI
CONNECTED DEVICES – VIRTUAL WORLD – VOLUMETRIC VIDEO – CODECS – PROTOCOLS – AUDIO**



At the intersection of Media and Connectivity

-  **Fast tech evolution**
-  **Global ecosystem**
-  **Built better together**
-  **Solution to pain-points**



Connected Devices
 Streaming Platforms
 Next Generation Codecs
 Internet-based Protocols
 eXtended Reality
 Wireless Video
 Real-Time Comms
 Data and Analytics
 Content Production
 Satellite
 Network APIs
 Artificial Intelligence
 Quality of Service
 Latency
 Energy Consumption
 Synchronization
 Spatial Media

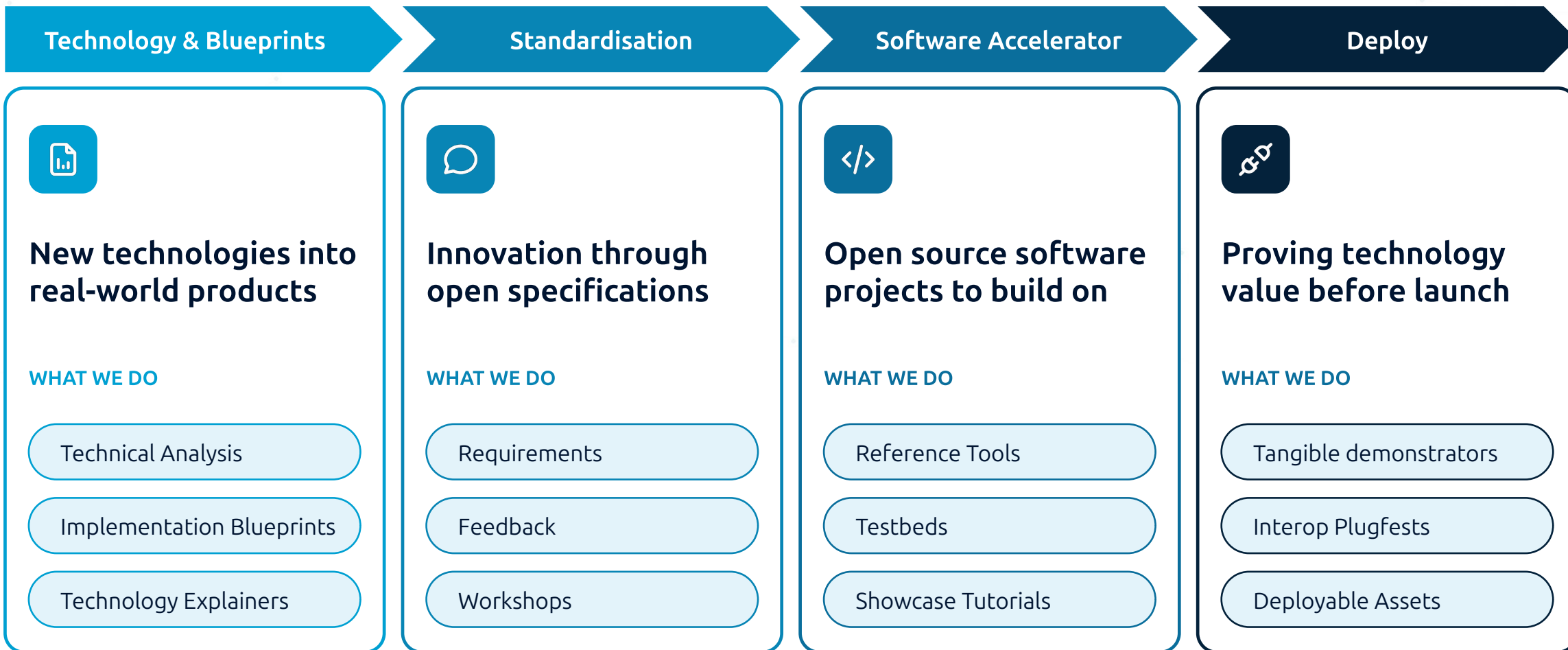
From open specifications to real-world deployments

SPECIFICATIONS DRIVING
MEDIA + CONNECTIVITY

IMPLEMENTED AND
PUSHED INTO ACTION



From open specifications to real-world deployments



19 projects across 7 technology areas



Content Delivery and Streaming

- 5G Media Streaming (5GMS)
- UE Data Collection, Reporting and Event Exposure
- Video Operation Points
- DVB-I Services over 5G Systems
- Content Delivery Protocols



Immersive Media

- XR Media & 3D Scenes
- Volumetric Video Experiences
- Dynamic Mesh Coding
- Beyond 2D Video Experiences
- Conversational Avatar Communication



Broadcast and Multicast

- 5G Broadcast - TV and Radio Services
- Public Warning Systems
- 5G Multicast Broadcast Services (MBS)



Real-Time Communications (RTC)

- Real-time Media Communication (RTC) Architecture



Non-Terrestrial Networks (NTN)

- Non-Terrestrial Networks in 5G Systems



Connected Media Production

- Network APIs for Connectivity Quality
- Non-Public Networks
- Time-Sensitive Communications (TSC)



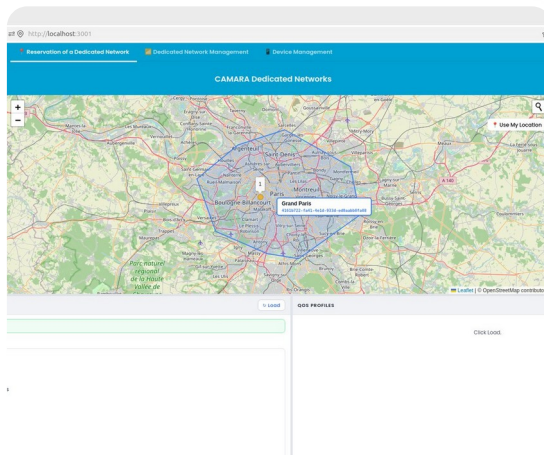
Towards 6G Media

- AI in Media Services

16 Reference Tools, **4** Testbeds across **65** Repositories

WHAT WE BUILD

The technology in action, in numbers



19

projects across 7 technology areas

16

Reference Tools, open-source software

4

Testbeds and evaluation frameworks

65

repositories: 45 public, 20 early access

201

issues raised on specifications

35

inputs and liaison statements to SDOs

JOIN THE EFFORT

What you get that you won't get alone

We offer a neutral framework to collaborate within minimal bureaucracy, focused on results.



Shape standards early

Shape the standards and technologies before others set them without you.



Early access to pre-public code

Access implementations before publication and be faster to market.



Prepare for plugfests & interop

Test your implementation in advance and build reference vector traces.



Mutualised effort to grow your project

From idea to reference code to demo and trial, on shared effort.



Exposure at events, demos and trials

Build once, together, instead of funding the same work alone.



De-risk on deployments

Back technologies that get deployed.



Going deeper

Technology, Standardisation, Software Accelerator, Deploy, Membership

Technology & Blueprints

Understand it. Prove its value. Scale it.



Specification analysis for
real-world applications



Making standards
actionable



Lowering the barrier to
entry



Technical Analysis

Understanding how
specifications support
application requirements.



Implementation Blueprints

Turn standards into
implementation-ready
blueprints.

Examples of **Standards Development Organizations (SDO)**



Standardisation Activities

Advancing standardization with implementation and experience-based feedback



Active engagement with standards bodies



Requirements based on application needs



Identification of bugs and enhancements



Requirements towards SDOs

Requirements towards standards bodies, backed by industry inputs and surveys.



Industry Surveys

Share your input to help shape the requirements 5G-MAG brings to standards bodies.



Feedback to SDOs

Per-SDO feedback tracked on GitHub, and how an issue is carried through to a correction or liaison input.



Liaison Statements & Inputs

Tables of LS sent to and received from SDOs.



Workshops for Standards

The workshops that fed directly into 3GPP study and work items.

Standardisation activities in numbers

201

issues raised on specifications through GitHub

3GPP

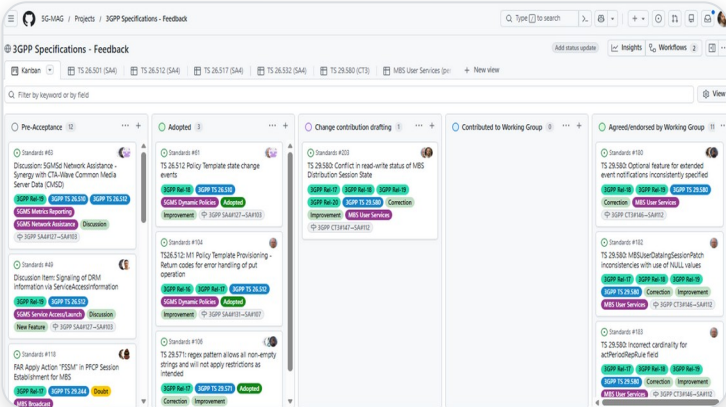
engagement as Market Representation Partner

35

inputs and liaison statements to SDOs

6

presentations at 3GPP requirements workshops



Issues found on specifications through GitHub



Profiles of Standards Specifications



Requirements towards Standards Bodies

Media Connectivity Software Accelerator



Early feedback and
validation



Software projects
& Dev community



No duplication and
fast deployment



Open by design
and IPR-friendly



Reference Tools

Standards, turned into
open code anyone can
build on.



Testbeds & Evaluation Frameworks

Reproducible test
environments and
benchmark frameworks.



Showcase Tutorials

Use-case driven
implementations towards
real-world applications.

Enabling open-source software from the start of every project

TODAY: OPEN SOURCE AFTER THE FREEZE

WITH OPEN SOURCE FROM THE START

Evaluation

Duplicated prototypes. Each organization builds its own to evaluate a draft; results differ and feedback is slow.

One shared baseline. Drafts are validated against running code before the freeze, while gaps are cheap to fix.

Interoperability

Late interop bugs. Testing starts from scratch at plugfests; errors, some in the specification, surface late.

Faster plugfests. Implementers start from the same code: shorter preparation, faster root-cause analysis.

Deployment

Slow deployment. Years can pass from a frozen specification to production-ready code; early adopters carry the cost alone.

Earlier deployment. End-to-end demos and trials run before commercial products, and feed evidence back into the standard.

1 Plan it at launch

2 Engage implementers during drafting

3 Make the value tangible

“Open-source software and evaluation platform: home and scope identified at launch.”

Software Accelerator in numbers

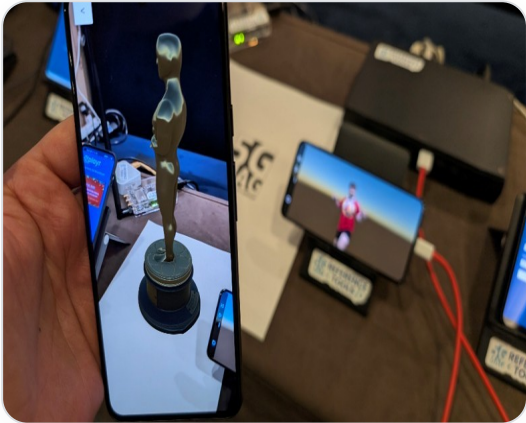
19

projects across media
and connectivity



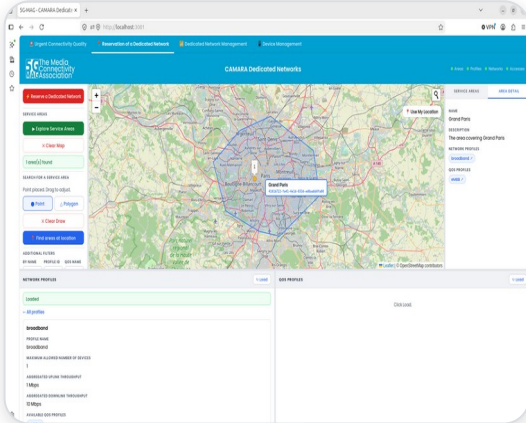
65

project repositories,
public and early access



Monthly

Dev Public Call and
rolling releases



Growing community of developers

Organizations contributing to the Software Accelerator



BBC



Big Blue Marble



Bitstem



Dolby Laboratories



EBU



Ericsson



Fraunhofer FOKUS



Huawei



idTOLU



InterDigital



iTEAM



Motion Spell



Nokia



Philips



phine.tech



Qualcomm



Resillion



Sofia Digital



Universitat Politècnica de València

A note on the Open-Source Software Licenses

Protecting IPR. Enabling industry collaboration.

What is the License Model?

1. The 5G-MAG Public License v1.0 is a modified version of Apache 2.0 which allows:
 - **Free Non-Commercial Use:** use, copy, modify and distribute for study, academic research, testing and validation.
 - **Commercial Exploitation:** use the software in commercial services, trials and live deployments, but:
 - **FRAND Patent Licensing:** essential contributor patents are made available under **FRAND** (Fair, Reasonable, and Non-Discriminatory) terms.
2. Repositories with software derived from other licenses keep those licenses.

How to Contribute?

1. Sign an Individual or Corporate Contributor License Agreement (CLA).
2. Anybody can contribute: no 5G-MAG membership needed.
3. Members get priority support from the Project Office, and define where resources go.



Deploy: demos, interop plugfests and applications

Specifications to code. Code to deployments.



Make value proposition tangible to the industry



Interoperability through early testing and plugfests



Supporting application development and trials



End-to-End Demos and Use Cases

User-facing demonstrations that abstract technical complexity.



Interoperability Events and Plugfests

With shared code, so interop does not start from scratch at plugfests.



Showcase at Industry Events and Trials

Promotion and demonstration of value, not just specs.

Where to meet us



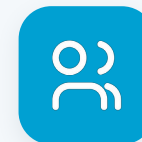
MWC Barcelona

Recurrent large event in Barcelona, with a 5G-MAG stand.



IBC

Recurrent large event in Amsterdam, with Reference Tools demos.



Future Media Townhall

Yearly conference at IBC, co-organized by 5G-MAG and IEEE BTS.



OSCAR Workshop

Open-Source Core, Applications and RAN.



OSMART Workshops

Open-Source Media Application Reference Tools.



Dev Public Call

Open monthly session, no membership needed. Last Friday of the month, 13:00 CET/CEST.



www.5g-mag.com/membership

JOIN THE EFFORT

Addressing industry pain points with open specifications

Member-driven

we add resources to
their efforts

Open

to the industry to
engage, collaborate

Agile

at the pace of
standards and market

Pragmatic

with actionable
activities & outcomes

3

WORKGROUPS focused on **TECHNOLOGY AND STANDARDS,**
DEVELOPMENT AND ECOSYSTEM, and **PROMOTION AND COMMUNICATION**

JOIN THE EFFORT

Our current membership



Membership information and Fee Structure

Membership Fee Category LARGE	Annual Revenue > €1 billion	Annual Membership Fee € 15,000
Membership Fee Category MEDIUM	Annual Revenue €100 million – €1 billion	Annual Membership Fee € 10,000
Membership Fee Category SMALL	Annual Revenue €5 million – €100 million	Annual Membership Fee € 5,000
Membership Fee Category MICRO	Annual Revenue < €5 million	Annual Membership Fee € 2,000
Micro also includes universities, regulators, public research bodies, institutions, NGOs, and non-profit organizations.		

Membership information, sponsorship opportunities, partnerships, questions on fees?
Contact **Eva Markvoort** at **markvoort@5g-mag.com**

JOIN THE EFFORT

Get involved

Ways to work with us, member or not



Bring your topic or project

Don't see your topic here? Bring it, or your project, as a member.



Early Access

Request access to Reference Tools and Testbeds still in development.



Contribute code

No membership needed: sign the CLA and open a pull request on GitHub.



Join the developers

The Dev Community on Slack and the Reference Tools mailing list.



Stay informed

Email updates, the 5G-MAG Magazine and The Handshake podcast.



Follow us

News on LinkedIn; talks and demos on YouTube.



Join the Effort

Help shape what connected media looks like next and scale your projects with the industry.

Become a member →

www.5g-mag.com/membership



GitHub



5g-mag.com